

2007
TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 2

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using a black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown for every question.
- Begin each question on a fresh sheet of paper.

Total marks (120)

- Attempt Questions 1-8
- All questions are of equal value

Total Marks – 120**Attempt Questions 1-8****All Questions are of equal value**

Begin each question on a NEW SHEET of paper, writing your name and question number at the top of the page. Extra paper is available.

Question 1

(15 marks)

Use a SEPARATE sheet of paper.

Marks

a) Find $\int \frac{e^{\tan x}}{\cos^2 x} dx$

2

b) i) Use partial fractions to evaluate

$$\int_0^1 \frac{5 dt}{(2t+1)(2-t)}$$

3

ii) Hence, and by using the substitution $t = \tan \frac{\theta}{2}$, evaluate

$$\int_0^{\frac{\pi}{2}} \frac{d\theta}{3 \sin \theta + 4 \cos \theta}$$

3

c) By using the table of standard integrals and manipulation, find

$$\int_0^1 \frac{dx}{\sqrt{4x^2 + 36}}$$

2

d) If $I = \int e^x \sin x dx$, find I .

3

e) By completing the square find

$$\int \frac{dx}{\sqrt{1-4x-x^2}}$$

2

End of Question 1

Question 2 (15 marks)

Use a SEPARATE sheet of paper.

Marks

a) Given $z_1 = i\sqrt{2}$ and $z_2 = \frac{2}{1-i}$

i) Express z_1 and z_2 in Mod / Arg form.

2

ii) If $z_1 = \omega z_2$, find the complex number ω in Mod / Arg form.

1

iii) α) On the Argand diagram plot the points P and Q representing the complex numbers z_1 and z_2 respectively.

1

 β) Show how to construct the point R representing $z_1 + z_2$.

1

iv) α) Find $\arg(z_1 + z_2)$

1

 β) Find the exact value of $\tan \frac{3\pi}{8}$

1

b) Draw a diagram to illustrate the locus of points z in the complex plane such that

i) $\operatorname{Arg}\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$

2

ii) $\operatorname{Re}(z) \leq 1$ and $|z - 3 + 4i| \leq 5$

2

iii) $|z - 3| + |z + 1| = 6$

2

c) Given that $(x-2)$ is a factor of $x^3 - 4x^2 + 7x - 6$ reduce $x^3 - 4x^2 + 7x - 6$ to irreducible factors over the complex field.

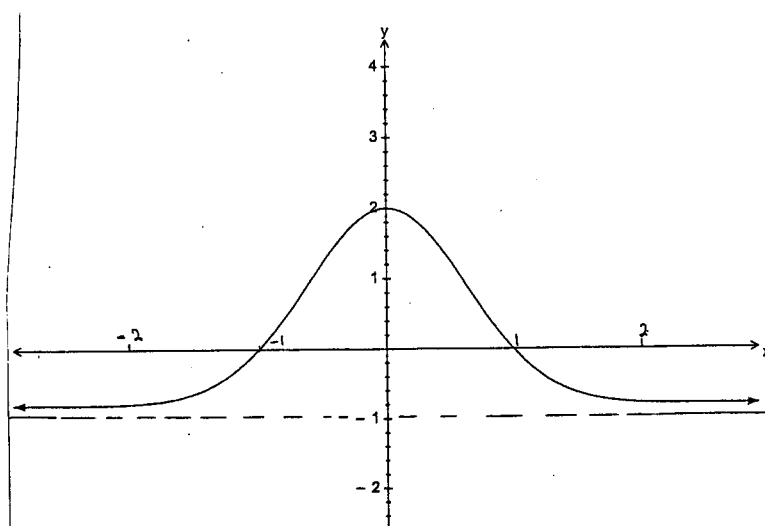
2

End of Question 2

Question 3 (15 marks) Use a SEPARATE sheet of paper.

Marks

- a) The sketch below is the even function $y = f(x)$.



On separate diagrams sketch each of the following, clearly showing all important features

(Each of the graphs for the questions below to be done on the sheets with the graph of $y = f(x)$ provided)

- | | |
|---|---|
| i) $y = f(x) - 2$ | 1 |
| ii) $y = f(x - 2)$ | ① |
| iii) $y = f(x) $ | 1 |
| iv) $y = [f(x)]^2$ | 2 |
| v) $y = \frac{1}{f(x)}$ | 2 |
| vi) $y^2 = f(x)$
$y = \sqrt{f(x)}$ | ② |
| b) Nine people gather to play football by forming two teams of four to play each other with the remaining person to be the referee. | |
| i) In how many ways can the teams be formed | ① |
| ii) If two particular people are not to be in the same team, how many ways are there then to choose the teams | ② |
| c) A particle moving in Simple Harmonic Motion has a speed of $10\sqrt{3} \text{ ms}^{-1}$ at the centre of its motion. Find its speed when it is at half of its amplitude. | |
| | ③ |

End of Question 3

Question 4

(15 marks)

Use a SEPARATE sheet of paper.

Marks

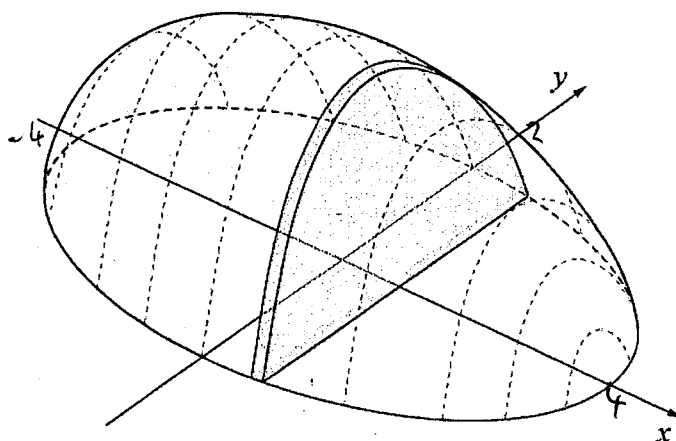
- a) Show that the area enclosed between the parabola $x^2 = 4ay$ and its latus rectum is $\frac{8a^2}{3}$ units².

3

(The latus rectum is the focal chord perpendicular to the axis of the parabola)

- b) A solid figure has as its base, in the xy plane, the ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$.

Cross-sections perpendicular to the x -axis are parabolas with latus rectums in the xy plane.



- i) Show that the area of the cross-section at $x = h$ is $\frac{16-h^2}{6}$ units².
[Use your answer to part (a)]

4

- ii) Hence, find the volume of this solid.

2

- c) i) Prove that $\int_0^a f(x)dx = \int_0^a f(a-x)dx$, (Let $u = a-x$)

2

- ii) Consider $f(x) = \frac{1}{1+\tan x}$ where $0 \leq x \leq \frac{\pi}{2}$ and $f\left(\frac{\pi}{2}\right) = 0$

2

$$\text{show that } f(x) + f\left(\frac{\pi}{2} - x\right) = 1$$

- iii) Hence evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{1+\tan x} dx$

2**End of Question 4**

Question 5

(15 marks)

Use a SEPARATE sheet of paper.

Marks

- a) i) On the same diagram sketch the graphs of the ellipses $E_1: \frac{x^2}{4} + \frac{y^2}{3} = 1$

and $E_2: \frac{x^2}{16} + \frac{y^2}{12} = 1$, showing clearly the intercepts on the axes. Show

the coordinates of the foci and the equations of the directrices of the ellipse E_1 .

5

- ii) $P(2 \cos p, \sqrt{3} \sin p)$ where $0 < p < \frac{\pi}{2}$, is a point on the ellipse E_1 . Use

(3)

differentiation to show that the tangent to the ellipse E_1 at P has equation

$$\frac{x \cos p}{2} + \frac{y \sin p}{\sqrt{3}} = 1.$$

- iii) The tangent to the ellipse E_1 at P meets the ellipse E_2 at the points

(2)

$Q(4 \cos q, 2\sqrt{3} \sin q)$ and $R(4 \cos r, 2\sqrt{3} \sin r)$, where $-\pi < q < \pi$

and $-\pi < r < \pi$. Show that q and r differ by $\frac{2\pi}{3}$.

- b) Let $P(x) = x^4 - 5x + 2$. The equation $P(x) = 0$ has roots α, β, γ and δ .

- i) Evaluate $P(0)$ & $P(1)$. Hence show that the equation $x^4 - 5x + 2 = 0$ has a real root between $x = 0$ and $x = 1$

1

- ii) Find the monic equation with roots $\alpha^2, \beta^2, \gamma^2$ and δ^2 . Hence or otherwise show that $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 0$.

2

- iii) Find the number of non-real roots of $x^4 - 5x + 2 = 0$, giving full reasons for your answer.

2

End of Question 5

Question 6 (15 marks) Use a SEPARATE sheet of paper.

Marks

a) i) Given $z = \cos \theta + i \sin \theta$, show that $z^n + z^{-n} = 2 \cos n\theta$.

2

ii) Write down $(z + z^{-1})$ in terms of $\cos n\theta$

1

iii) Show that $(z + z^{-1})^4 = z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4}$

1

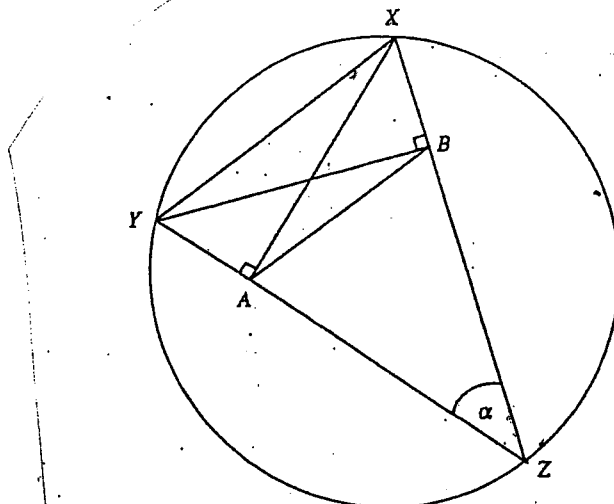
iv) Hence, express $\cos^4 \theta$ in terms of $\cos n\theta$.

(3)

v) Using your answer to part (ii), evaluate $\int_0^{\pi/4} \cos^4 \theta d\theta$.

(2)

b)



XY is a fixed chord of a circle. Z is a point on the major arc XY. The perpendicular from X to the chord YZ meets YZ at A. The perpendicular from Y to the chord XZ meets XZ at B. $\angle XZY = \alpha$

i) Copy the diagram

ii) Show that ABXY is a cyclic quadrilateral

1

iii) Show that $\triangle ABZ \sim \triangle XYZ$.

2

iv) Hence show that $AB = XY \cos \alpha$.

2

v) Deduce that as Z moves on the major arc XY, the length of AB is constant.

1

End of Question 6

Question 7

(15 marks)

Use a SEPARATE sheet of paper.

Marks

- a) A mass of m kg is allowed to fall under gravity from a stationary position h metres above the ground. It experiences resistance proportional to the square of its velocity, $v \text{ ms}^{-1}$.

i) Explain why the equation for this motion is $\ddot{x} = g - kv^2$,
(where k is a constant). 1

ii) Show that $\ddot{x} = v \frac{dv}{dx}$. 1

iii) Hence show that $v^2 = \frac{g}{k}(1 - e^{-2kx})$. 4

iv) Find the velocity at which the mass hits the ground in terms of g , h and k . 1

v) Calculate the terminal velocity of the object, given that h is sufficiently large to allow terminal velocity to be reached. 1

b) i) On the same axes sketch the graphs of $y = \sqrt{1-x^2}$ and $y = \frac{1}{\sqrt{1-x^2}}$. 2

ii) The region bounded by the curve $y = \frac{1}{\sqrt{1-x^2}}$, the coordinate 3

axes and the line $x = \frac{1}{2}$ is rotated through one complete revolution about the line $x = 6$. Use the method of cylindrical shells to show that volume $V \text{ units}^3$ of the solid of revolution is given by

$$V = 2\pi \int_0^{1/2} \frac{6-x}{\sqrt{1-x^2}} dx.$$

iii) Hence find the value of V in simplest exact form. 2

End of Question 7

Question 8

(15 marks)

Use a SEPARATE sheet of paper.

Marks

- a) i) If $I_n = \int_0^1 (x^2 - 1)^n dx$, $n = 0, 1, 2, \dots$ show that

4

$$I_n = \frac{-2n}{2n+1} I_{n-1}, \quad n = 1, 2, 3, \dots$$

- ii) Hence use the method of Mathematical Induction to show that

4

$$I_n = \frac{(-1)^n 2^{2n} (n!)^2}{(2n+1)!}$$

for all positive integers n

- b) A polynomial $P(x)$ is divided by $x^2 - a^2$ where $a \neq 0$ and the remainder is $px + q$.

ie $P(x) = (x^2 - a^2) \cdot Q(x) + px + q$ for some polynomial $Q(x)$

- i) Show that $p = \frac{1}{2a} \{P(a) - P(-a)\}$ and $q = \frac{1}{2} \{P(a) + P(-a)\}$

3

- ii) Find the remainder when the polynomial $P(x) = x^n - a^n$ is divided

by $x^2 - a^2$ for the cases

$\alpha)$ n even

2

$\beta)$ n odd

2

End of Examination

$$\begin{aligned} \text{1a } \int \frac{e^{\tan x}}{\cos^2 x} dx \\ = \int e^u du \\ = e^u + c \\ = e^{\tan x} + c \end{aligned}$$

$$\begin{aligned} \text{let } u = \tan x \\ du = \sec^2 x dx \end{aligned}$$

$$\text{b (i) } \frac{5}{(2t+1)(2-t)} = \frac{A}{2t+1} + \frac{B}{2-t} \quad \text{where } A=2, B=1$$

by Heaviside's rule

$$\begin{aligned} \therefore \int_0^1 \frac{5 dt}{(2t+1)(2-t)} &= \int_0^1 \left(\frac{2}{2t+1} + \frac{1}{2-t} \right) dt \\ &= \left[\ln(2t+1) - \ln(2-t) \right]_0^1 \\ &= \ln 2 \end{aligned}$$

$$\text{(ii) } \int_0^{\frac{\pi}{2}} \frac{2d\theta}{3\sin\theta + 4\cos\theta} = \int_0^1 \frac{\frac{2dt}{1+t^2}}{\frac{6t}{1+t^2} + \frac{4-4t^2}{1+t^2}}$$

$$t = \tan \frac{\theta}{2}$$

$$dt = \frac{2dt}{1+t^2}$$

$$\sin \theta = \frac{2t}{1+t^2}$$

$$\cos \theta = \frac{1-t^2}{1+t^2}$$

$$\theta = \frac{\pi}{2} \Rightarrow t = \tan \frac{\pi}{4} = 1$$

$$\theta = 0 \Rightarrow t = \tan 0 = 0$$

$$\begin{aligned} &= \int_0^1 \frac{2 dt}{6t + 4 - 4t^2} \\ &= \int_0^1 \frac{dt}{(2t+1)(t-2)} \\ &= \frac{1}{5} \ln 2 \end{aligned}$$

$$\begin{aligned} \text{c } \int_0^1 \frac{dx}{\sqrt{4x^2+36}} &= \frac{1}{2} \int_0^1 \frac{dx}{\sqrt{x^2+9}} \\ &= \frac{1}{2} \left[\ln(x + \sqrt{x^2+9}) \right]_0^1 \\ &= \frac{1}{2} \ln\left(\frac{1+\sqrt{10}}{3}\right) \end{aligned}$$

$$\text{d } I = \int e^x \sin x dx = -e^x \cos x + \int e^x \cos x dx$$

$$u = e^x \quad v = -\cos x$$

$$u' = e^x \quad v' = \sin x$$

$$= -e^x \cos x + e^x \sin x - \int e^x \sin x dx$$

$$\therefore 2I = e^x (\sin x - \cos x)$$

$$I = \frac{1}{2} e^x (\sin x - \cos x) + c$$

$$\begin{aligned} \text{e } \int \frac{dx}{\sqrt{1-4x-x^2}} &= \int \frac{dx}{1+4-(x^2+4x+4)} \\ &= \int \frac{dx}{5-(x+2)^2} \\ &= \sin^{-1}\left(\frac{x+2}{\sqrt{5}}\right) + c \end{aligned}$$

2a (i) $z_1 = \sqrt{2} \cos \frac{\pi}{4}$

$$z_2 = \frac{2}{(1-i)} \cdot \frac{(1+i)}{(1+i)}$$

$$= 1+i$$

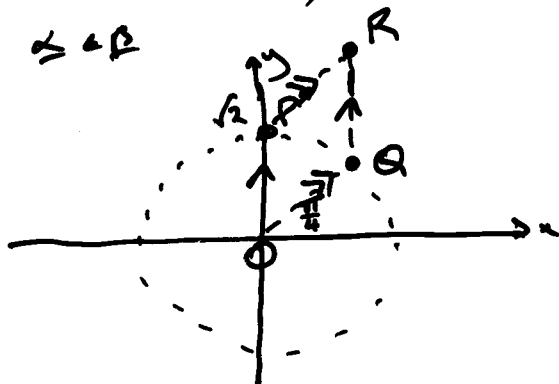
$$= \sqrt{2} \cos \frac{\pi}{4}$$

(ii) $w = \frac{z_1}{z_2}$

$$= \frac{\sqrt{2} \cos \frac{\pi}{4}}{\sqrt{2} \cos \frac{\pi}{4}}$$

$$= \cos(-\frac{\pi}{4})$$

(iii) $\angle \alpha$



$P + Q$

R

(iv) $\therefore OPRQ$ is a rhombus

$$\therefore \arg(z_1 + z_2) = \angle xOR$$

$$= \angle xOQ + \angle QOR$$

$$= \frac{\pi}{4} + \frac{\pi}{8} \quad (\text{diag. of rhombus bisects angle})$$

$$= \frac{3\pi}{8}$$

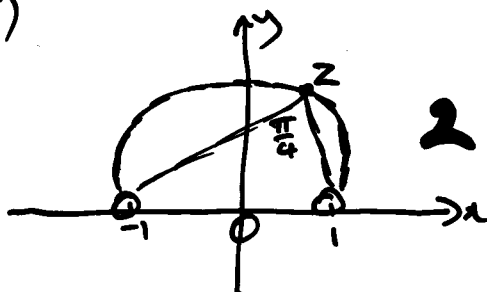
$\therefore z_1 + z_2 = i\sqrt{2} + 1+i$

$$= 1 + (1+\sqrt{2})i$$

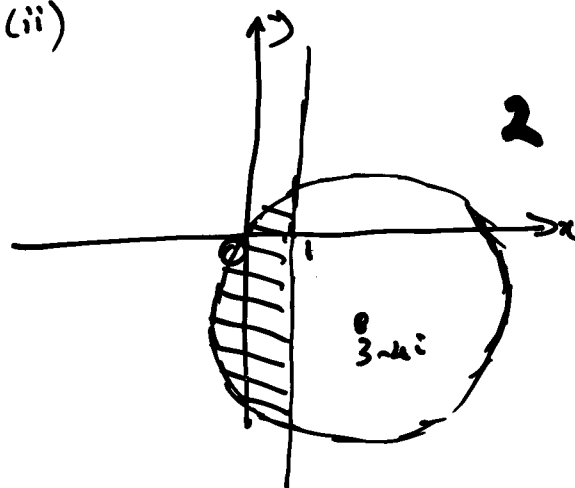
$\therefore \tan \frac{3\pi}{8} = \frac{1+\sqrt{2}}{1}$

$$= 1+\sqrt{2}$$

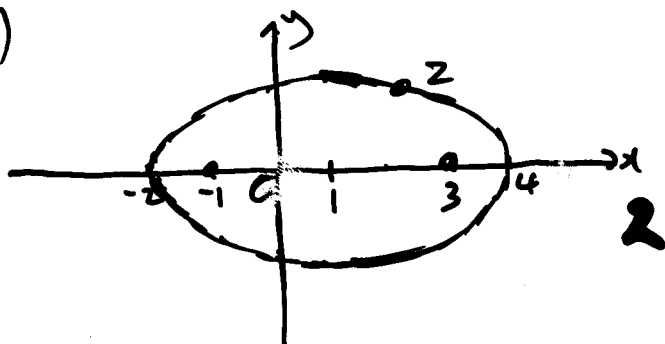
b (i)



(ii)



(iii)



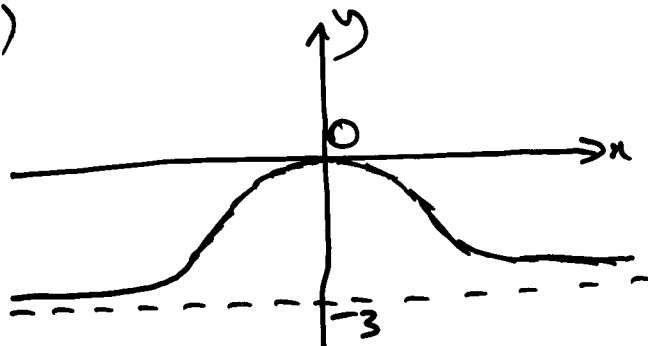
Let $P(x) = x^3 - 4x^2 + 7x - 6$

$$\begin{array}{r} x^2 - 2x + 3 \\ x-2 \overline{) x^3 - 4x^2 + 7x - 6} \\ \underline{x^3 - 2x^2} \\ -2x^2 + 7x \\ \underline{-2x^2 + 4x} \\ 3x - 6 \end{array}$$

$$\begin{aligned} \therefore P(x) &= (x-2)(x^2 - 2x + 3) \\ &= (x-2)(x^2 - 2x + 1 + 2) \\ &= (x-2)((x-1)^2 + 2) \\ &= (x-2)[(x-1)^2 - (\sqrt{2}i)^2] \\ &= (x-2)(x-1-\sqrt{2}i)(x-1+\sqrt{2}i) \end{aligned}$$

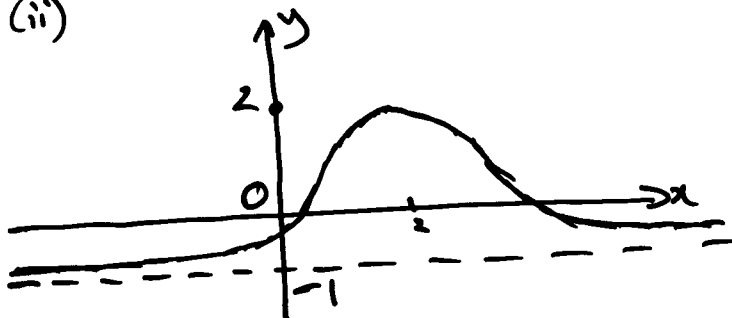
2

3a (i)



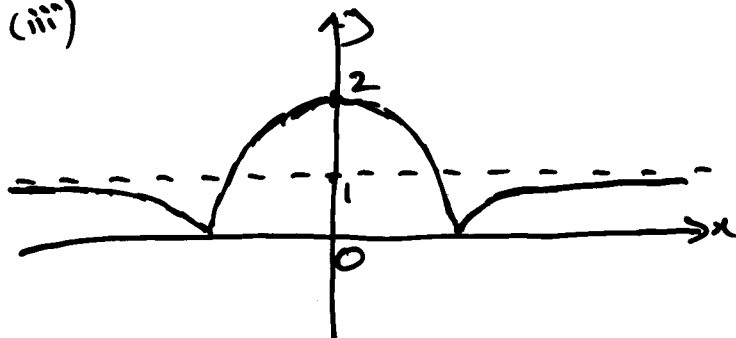
$$y = f(x) - 2$$

(ii)



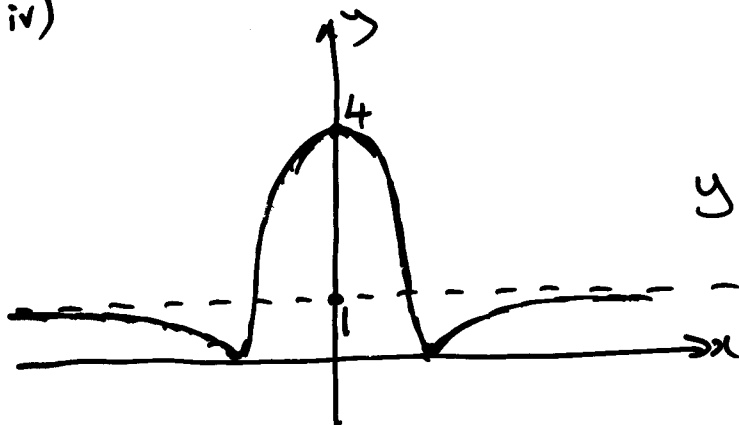
$$y = f(x-2)$$

(iii)



$$y = |f(x)|$$

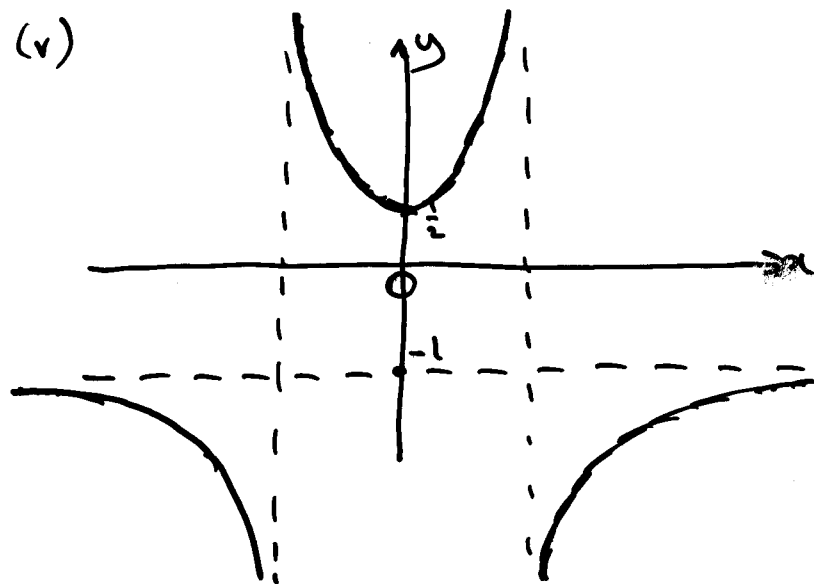
(iv)



$$y = [f(x)]^2$$

2

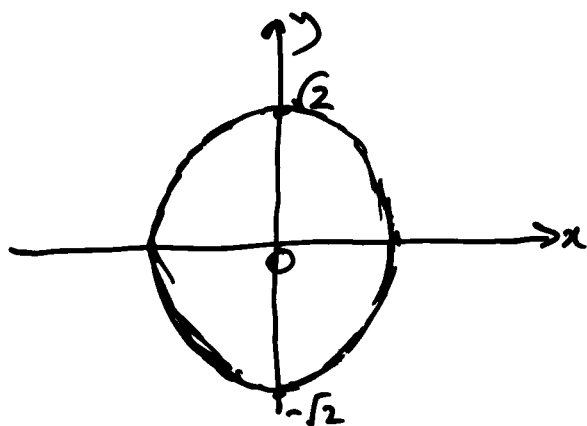
(v)



$$y = f(x)$$

2

(vi)



$$y^2 = f(x)$$

2

1 (i) $\frac{\text{Choose referee} \times \text{choose 4 from 8}}{2} = \frac{{}^8C_1 \times {}^8C_4}{2}$

$$= 315$$

1

(ii) Let A, B be the two particular people

$$\# \text{ ways} = (\# \text{ ways with A or B ref}) + (\# \text{ ways with A, B playing})$$

$$= \frac{2 \times {}^8C_4}{2}$$

$$+ 7 \times (\text{team with A in, B not}) \times (\text{team with B in, A not})$$

$$= {}^8C_4$$

$$+ 7 \times {}^6C_3$$

$$= 210$$

2

= Set up S.H.M. with O at centre of motion

$$\therefore v^2 = n^2(a^2 - x^2)$$

$$\text{When } x=0, v=10\sqrt{3}$$

$$\text{When } x = \frac{a}{2}$$

$$\Rightarrow 300 = n^2 a^2$$

$$v^2 = n^2 \left(a^2 - \frac{a^2}{4} \right)$$

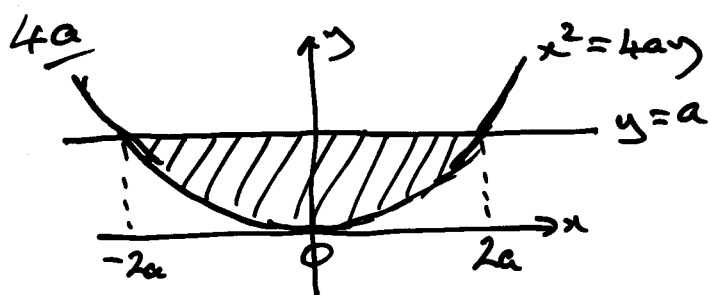
$$= \frac{3n^2 a^2}{4}$$

$$= \frac{3}{4} \times 300$$

$$= 225$$

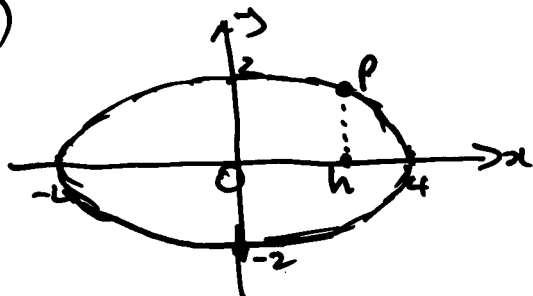
$$\therefore v = 15 \text{ ms}^{-1}$$

3



$$\begin{aligned}
 A &= 2 \int_0^{2a} \left(a - \frac{x^2}{4a} \right) dx \\
 &= 2 \left[ax - \frac{x^3}{12a} \right]_0^{2a} \\
 &= 2 \left[2a^2 - \frac{8a^3}{12a} - (0-0) \right] \\
 &= \frac{8a^2}{3}
 \end{aligned}$$

6 (i)



Take cross section through point P on ellipse where $x = h$.

$$\begin{aligned}
 \therefore \frac{h^2}{16} + \frac{y^2}{4} &= 1 \\
 h^2 + 4y^2 &= 16 \\
 y^2 &= \frac{16-h^2}{4} \\
 y &= \pm \frac{\sqrt{16-h^2}}{2}
 \end{aligned}$$

$\therefore$ length of latus rectum is $2y = \sqrt{16-h^2}$

But length of latus rectum is $4a$ $\therefore a = \frac{\sqrt{16-h^2}}{4}$

From a , area of parabolic cross section is $\frac{8}{3} \left(\frac{\sqrt{16-h^2}}{4} \right)^2$
 $= \frac{8}{3} \left(\frac{16-h^2}{16} \right)$
 $= \frac{16-h^2}{6}$

(ii) Vol. of slice $\delta V = \frac{16-h^2}{6} \cdot \delta h$

Total vol $= 2 \times \lim_{\delta h \rightarrow 0} \sum_{h=0}^4 \frac{16-h^2}{6} \cdot \delta h$

$$\begin{aligned}
 &= 2 \int_0^4 \frac{16-h^2}{6} dh \\
 &= \frac{1}{3} \left[16h - \frac{h^3}{3} \right]_0^4 \\
 &= \frac{128}{9} \text{ cu. units}
 \end{aligned}$$

$$c_{-}(i) \int_0^a f(x) dx$$

$$= \int_a^0 f(a-u) x^{-1} du$$

$$\text{let } u = a - x \\ du = -dx$$

$$= \int_0^a f(a-u) du$$

$$\text{Limits } x=a \Rightarrow u=0 \\ x=0 \Rightarrow u=a$$

$$= \int_0^a f(a-x) dx$$

} 2

$$(ii) LHS = f(x) + f\left(\frac{\pi}{2} - x\right)$$

$$= \frac{1}{1+\tan x} + \frac{1}{1+\tan\left(\frac{\pi}{2} - x\right)}$$

$$= \frac{1}{1+\tan x} + \frac{1}{1+\cot x} \cdot \frac{\tan x}{\tan x}$$

$$= \frac{1}{1+\tan x} + \frac{\tan x}{\tan x + 1}$$

$$= \frac{1+\tan x}{1+\tan x}$$

$$= 1 \\ = RHS$$

} 2

$$(iii) \int_0^{\frac{\pi}{2}} \frac{dx}{1+\tan x} = \int_0^{\frac{\pi}{2}} \frac{dx}{1+\tan\left(\frac{\pi}{2} - x\right)} \quad \text{from (i)}$$

$$= \int_0^{\frac{\pi}{2}} \left(1 - \frac{1}{1+\tan x}\right) dx \quad \text{from (ii)}$$

$$= \int_0^{\frac{\pi}{2}} 1 dx - \int_0^{\frac{\pi}{2}} \frac{dx}{1+\tan x}$$

$$\therefore 2 \int_0^{\frac{\pi}{2}} \frac{dx}{1+\tan x} = \int_0^{\frac{\pi}{2}} 1 dx$$

$$\therefore \int_0^{\frac{\pi}{2}} \frac{dx}{1+\tan x} = \frac{1}{2} [x]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{4}$$

} 2

5a (i) $E_1: \frac{x^2}{4} + \frac{y^2}{3} = 1$

$a=2, b=\sqrt{3}$

$e = \frac{1}{2}$ - ①

Foci $(\pm 1, 0)$

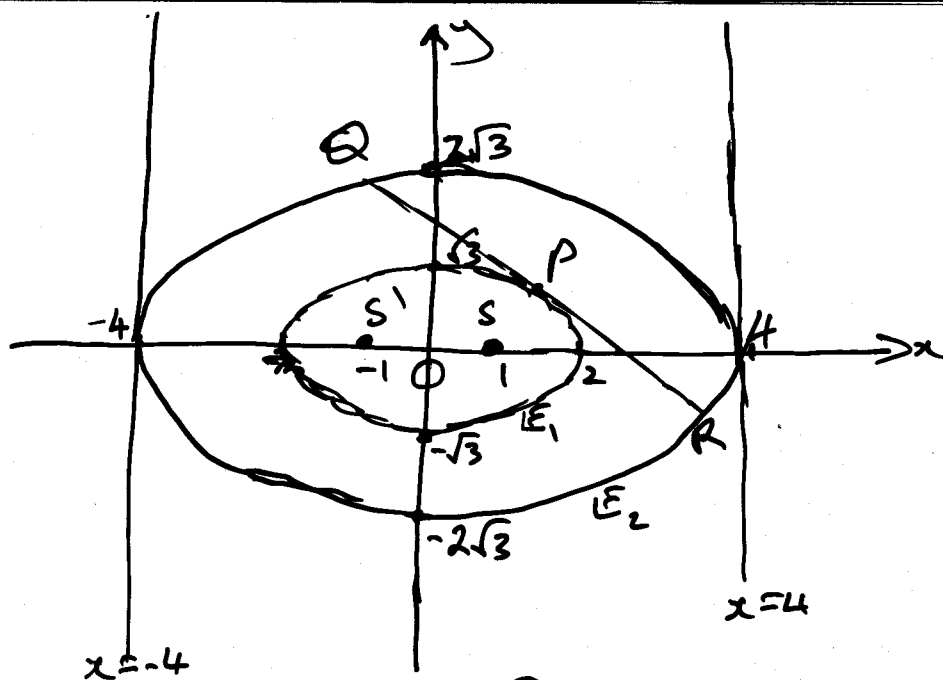
Directrices $x = \pm 4$

E_1 intercepts = 1

foci = 1

directrices = 1

E_2 intercepts = 1



(ii) At P: $x = 2 \cos p$
 $y = \sqrt{3} \sin p \Rightarrow \frac{dx}{dp} = -2 \sin p$
 $\frac{dy}{dp} = \sqrt{3} \cos p$ } OR implicit diffⁿ to get $\frac{dy}{dx} = \frac{-3x}{4y}$

At P: $\frac{dy}{dx} = \frac{dy}{dp} \cdot \frac{dp}{dx}$
 $\therefore$ grad. tangent at P $= \frac{\sqrt{3} \cos p}{-2 \sin p}$ } At P, grad. of tangent $= \frac{-3 \times 2 \cos p}{4 \times \sqrt{3} \sin p}$
 $= \frac{-\sqrt{3} \cos p}{2 \sin p}$

Eq'n tangent at P is $y - \sqrt{3} \sin p = \frac{\sqrt{3} \cos p}{-2 \sin p} (x - 2 \cos p)$
 etc }
 to get $\frac{x \cos p}{2} + \frac{y \sin p}{\sqrt{3}} = 1$

(iii) At Q, $x = 4 \cos q$
 $y = 2\sqrt{3} \sin q$ } satisfies $\frac{x \cos p}{2} + \frac{y \sin p}{\sqrt{3}} = 1$

$\therefore \frac{4 \cos q \cos p}{2} + \frac{2\sqrt{3} \sin q \sin p}{\sqrt{3}} = 1$

$\cos q \cos p + \sin q \sin p = \frac{1}{2}$

$\cos(q-p) = \frac{1}{2}$

$\cos(r-p) = \frac{1}{2}$

Similarly, at R,

Hence $q-p = \pm \frac{\pi}{3}$ and $r-p = \mp \frac{\pi}{3}$

(q, r are distinct values)

$\therefore (q-p) - (r-p) = \pm \frac{\pi}{3} - (\mp \frac{\pi}{3})$

$q-r = \pm \frac{2\pi}{3}$

ie q and r differ by $\frac{2\pi}{3}$

Q. (i) $P(0) = 2$
 $P(1) = -2$ } $P(0)$ and $P(1)$ are on opposite sides of x axis and $P(x)$ is continuous
 $\therefore P(x)$ has real root between $x=0$ & $x=1$ }

(ii) Eqn with roots $\alpha^2, \beta^2, \gamma^2, \delta^2$ is, $P(\sqrt{x}) = 0$

$$(\sqrt{x})^4 - 5\sqrt{x} + 2 = 0$$

$$x^2 + 2 = 5\sqrt{x}$$

$$x^4 + 4x^2 + 4 = 25x$$

$$x^4 + 4x^2 - 25x + 4 = 0 \quad -1$$

Using sum of roots $= -\frac{b}{a}$

$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 0 \quad -1$$

(iii) For the eqn $P(x) = 0$, with roots $\alpha, \beta, \gamma, \delta$

None of $\alpha, \beta, \gamma, \delta = 0$ as $x=0$ not a solution
 and one root, say α , is real (from part (i)) }

$$\therefore \alpha^2 > 0 \quad \& \quad \beta^2 + \gamma^2 + \delta^2 < 0$$

ie at least one of β, γ, δ is not real

But coefficients of $P(x)$ are real

$\therefore$ non-real roots as conjugate pairs }

$\therefore$ 2 non-real roots.

6a (i) $z = \cos \theta + i \sin \theta$

$z^n = \cos n\theta + i \sin n\theta$

$z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$

$= \cos n\theta - i \sin n\theta$ } by De Moivre

as $\cos$ even function
 $\sin$ odd function.

$\therefore z^n + z^{-n} = 2 \cos n\theta$

(ii) Let $n=1$ $\therefore z + z^{-1} = 2 \cos \theta$

(iii) Expand $(z + z^{-1})^4$ to get $z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4}$
by Pascal's triangle or repeated expansion

(iv) From (i) with $n=4 \Rightarrow (z^4 + z^{-4}) = 2 \cos 4\theta$
with $n=2 \Rightarrow (z^2 + z^{-2}) = 2 \cos 2\theta$

Also $(z + z^{-1})^4 = (2 \cos \theta)^4$
 $= 16 \cos^4 \theta$

Using (iii) $16 \cos^4 \theta = (z^4 + z^{-4}) + 4(z^2 + z^{-2}) + 6$
 $= 2 \cos 4\theta + 8 \cos 2\theta + 6$

$\therefore \cos^4 \theta = \frac{1}{8} (\cos 4\theta + 4 \cos 2\theta + 3)$

(v) $\int_0^{\frac{\pi}{4}} \cos^4 \theta d\theta = \frac{1}{8} \int_0^{\frac{\pi}{4}} (\cos 4\theta + 4 \cos 2\theta + 3) d\theta$
 $= \frac{1}{8} \left[\frac{1}{4} \sin 4\theta + 2 \sin 2\theta + 3\theta \right]_0^{\frac{\pi}{4}}$

$= \frac{8 + 3\pi}{32}$

6 (ii) $ABXY$ is a cyclic quad. as interval XY subtends equal angles at A and B on same side of it. -1

OR (converse of angles at circumference)

(iii) Let $\angle BAX = \theta$

$\therefore \angle BAZ = 90 - \theta$ (straight angle)

$\angle BYX = \theta$ (angles at circumference)

$\angle BXY = 90 - \theta$ (angle sum of $\triangle BXY$) } 1

Thus in $\triangle$'s ABZ, XYZ

$\angle Z$ common

$\angle BAZ = \angle YXZ = 90 - \theta$

$\therefore \triangle ABZ \parallel \triangle XYZ$ (AAA) -1

(iv) In $\triangle BYZ$, $\cos \alpha = \frac{BZ}{YZ}$ } 1

But $\frac{BZ}{YZ} = \frac{AB}{XY}$ (corr. sides of sim $\triangle$'s) } 1

$\therefore \cos \alpha = \frac{AB}{XY}$ }

$AB = XY \cos \alpha$

(v) XY is a fixed chord i.e. XY is constant
 α is constant.

$\therefore AB$ is constant

7a (i) As P falls, forces are $\uparrow mkv^2$

$\downarrow mg$

Taking $\downarrow$ as positive resultant force on P is < 0 at point of release

$$m\ddot{x} = mg - mkv^2$$

$$\ddot{x} = g - kv^2$$

$$(ii) \ddot{x} = \frac{dv}{dt}$$
$$= \frac{dv}{dx} \cdot \frac{dx}{dt}$$
$$= v \frac{dv}{dx}$$

$$(iii) v \frac{dv}{dx} = g - kv^2$$

$$\frac{dv}{dx} = \frac{g - kv^2}{v}$$

$$\frac{dx}{dv} = \frac{v}{g - kv^2}$$

$$x = \int \frac{v}{g - kv^2} dv$$

$$x = -\frac{1}{2k} \ln(g - kv^2) + c$$

$$\text{When } x=0, v=0 \Rightarrow c = \frac{1}{2k} \ln g$$

$$\therefore x = -\frac{1}{2k} \ln(g - kv^2) + \frac{1}{2k} \ln g$$

$$-2kx = \ln\left(\frac{g - kv^2}{g}\right)$$

$$\frac{g - kv^2}{g} = e^{-2kx}$$

$$g - kv^2 = g e^{-2kx}$$

etc

$$v^2 = \frac{g}{k} (1 - e^{-2kx})$$

(iv) Hits ground when $x = h$

$$\therefore v^2 = \frac{g}{k} (1 - e^{-2kh})$$

$$v = \sqrt{\frac{g}{k} (1 - e^{-2kh})}$$

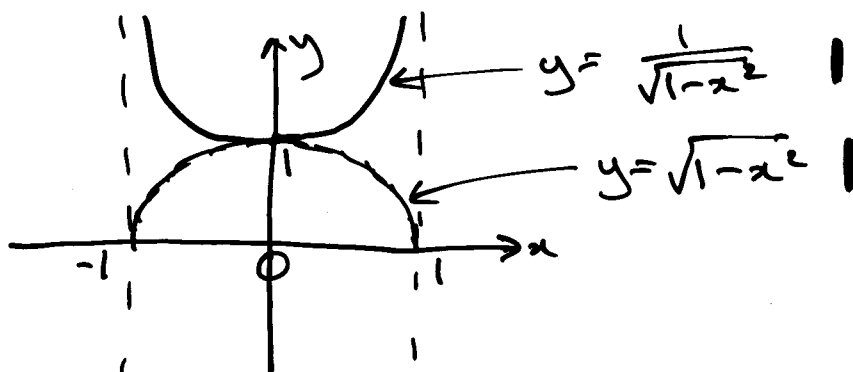
taking pos. square root as $\downarrow$ is pos.

(v) Terminal vel. when $\ddot{x} = 0$

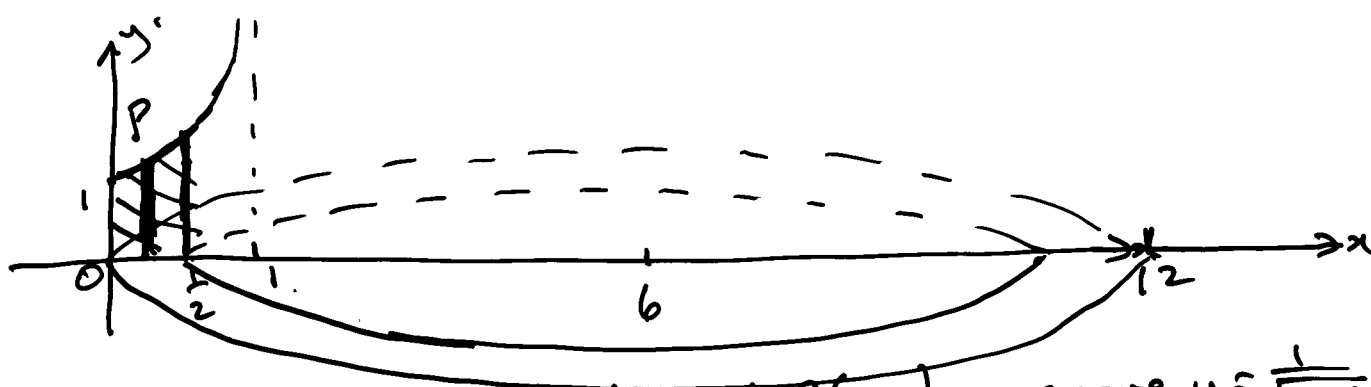
$$g - kv^2 = 0$$

$$v = \sqrt{\frac{g}{k}}$$

6 (i)



(ii)

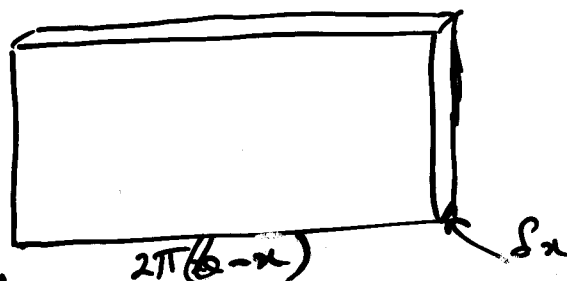


Take thin slice (vertical) through $P(x, y)$ on curve, $y = \frac{1}{\sqrt{1-x^2}}$ thickness δx , and rotate it about line $x=6$ to form a thin-walled hollow cylindrical shell of radius $6-x$.

When "opened" shell is

1 for "setup"

$$y = \frac{1}{\sqrt{1-x^2}}$$



$$\begin{aligned} \text{Vol. shell } \delta V &= 2\pi(6-x) \cdot \frac{1}{\sqrt{1-x^2}} \cdot \delta x \\ &= \frac{2\pi(6-x)}{\sqrt{1-x^2}} \delta x \end{aligned}$$

$$\begin{aligned} \text{Total vol. } V &= \lim_{\delta x \rightarrow 0} \sum_{x=0}^{\frac{1}{2}} \frac{2\pi(6-x)}{\sqrt{1-x^2}} \delta x \\ &= 2\pi \int_0^{\frac{1}{2}} \frac{6-x}{\sqrt{1-x^2}} dx \end{aligned}$$

$$\begin{aligned} \text{(iii) } V &= 12\pi \int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^2}} + \pi \int_0^{\frac{1}{2}} -2x(1-x^2)^{-\frac{1}{2}} dx \\ &= 12\pi [\sin^{-1}x]_0^{\frac{1}{2}} + \pi [2(1-x^2)^{\frac{1}{2}}]_0^{\frac{1}{2}} \end{aligned}$$

= etc

$$= 2\pi^2 + \pi\sqrt{3} - 2\pi \text{ cu. units.}$$

$$\begin{aligned} \text{since } \int f'(u)[f(u)]^n du &= \frac{[f(u)]^{n+1}}{n+1} \end{aligned}$$

8a (i) $I_n = \int_0^1 (x^2-1)^n dx$

$$\left. \begin{array}{l} u = (x^2-1)^n \quad v = x \\ u' = 2nx(x^2-1)^{n-1} \quad v' = 1 \end{array} \right\}$$

$$\begin{aligned} \therefore I_n &= \left[x(x^2-1)^n \right]_0^1 - 2n \int_0^1 x^2(x^2-1)^{n-1} dx \\ &= 0 - 2n \int_0^1 \left[(x^2-1)(x^2-1)^{n-1} + 1(x^2-1)^{n-1} \right] dx \\ &= -2n \int_0^1 (x^2-1)^n + (x^2-1)^{n-1} dx \\ &= -2n [I_n + I_{n-1}] \end{aligned}$$

$$\therefore (2n+1) I_n = -2n \cdot I_{n-1}$$

$$I_n = \frac{-2n}{(2n+1)} I_{n-1}$$

(ii) Prove true for $n=1$

By proposition $I_1 = \frac{(-1)^1 2^2 (1!)^2}{3!}$
 $= -\frac{2}{3}$

But $I_1 = \int_0^1 (x^2-1) dx$
 $= \left[\frac{x^3}{3} - x \right]_0^1$
 $= -\frac{2}{3}$

$\therefore$ proposition true for $n=1$

Assume true for $n=k$ (integer k)
 i.e. assume $I_k = \frac{(-1)^k 2^{2k} (k!)^2}{(2k+1)!}$

Prove true for $n=k+1$ if true for $n=k$

i.e. prove $I_{k+1} = \frac{(-1)^{k+1} 2^{2k+2} [(k+1)!]^2}{(2k+3)!}$

Now $I_{k+1} = \frac{-2(k+1)}{(2k+3)} \cdot I_k$ from part (i)

$$\begin{aligned} &= \frac{-2(k+1)}{(2k+3)} \cdot \frac{(-1)^k 2^{2k} (k!)^2}{(2k+1)!} \quad \text{by assumption true for } n=k \\ &= \frac{(-1)^{k+1} 2^{2k+1} (k+1)(k!)^2}{(2k+3)(2k+1)!} \cdot \frac{2(k+1)}{(2k+2)} \\ &= \frac{(-1)^{k+1} 2^{2k+2} [(k+1)!]^2}{(2k+3)!} \quad \text{+ Conclusion} \end{aligned}$$

Q (i) $P(x) = (x^2 - a^2)Q(x) + px + q$

So $P(a) = 0 \cdot Q(a) + pa + q$
 $\therefore pa + q = P(a) \dots \textcircled{1}$

Also $P(-a) = 0 \cdot Q(-a) - pa + q$
 $\therefore -pa + q = P(-a) \dots \textcircled{2}$

$\textcircled{1} + \textcircled{2} \quad 2q = P(a) + P(-a)$
 $q = \frac{1}{2} [P(a) + P(-a)] \quad -1$

$\textcircled{1} - \textcircled{2} \quad 2pa = P(a) - P(-a)$
 $p = \frac{1}{2a} [P(a) - P(-a)] \quad -1$

(ii) $P(x) = x^n - a^n$

When n even $P(a) = a^n - a^n$
 $= 0$

and $P(-a) = (-a)^n - a^n$
 $= a^n - a^n$ as n even
 $= 0$

$\therefore p = \frac{1}{2a} [0 - 0] = 0$ and $q = \frac{1}{2} [0 + 0] = 0$

$\therefore$ when n even, remainder $px + q$ becomes $0x + 0$
 i.e remainder is 0

When n odd $P(a) = a^n - a^n$
 $= 0$

and $P(-a) = (-a)^n - a^n$
 $= -a^n - a^n$ as n odd
 $= -2a^n$

~~$\therefore$ when n odd, remainder $px + q$ becomes~~

~~$\therefore p = \frac{1}{2a} [0 - (-2a^n)] = a^{n-1}$ and $q = \frac{1}{2} [0 + (-2a^n)] = -a^n$~~

$\therefore$ when n odd, remainder $px + q$ becomes
 $a^{n-1}x - a^n$